

JavaScript Beginner Practice Questions (Phase -2) Sheet - 4

Beginner Level

1. Create an Object

Create an object for a student with:

- name
- age
- course

Then print all values.

2. Access Properties

Given:

```
const car = {  
  brand: "BMW",  
  model: "M4",  
  year: 2022  
}
```

Print:

- brand
- model

using both:

- dot notation
 - bracket notation
-

3. Update Object Value

Change the age of a user from 20 to 25.

```
const user = {  
  name: "Anubhav",  
  age: 20  
}
```

4. Add New Property

Add a new property:

```
isAdmin: true
```

to this object.

5. Delete Property

Remove the `password` property from the object.

```
const account = {  
  username: "john",  
  password: "12345"  
}
```

Intermediate Level

6. Count Properties

Write a function that returns how many properties an object has.

Example:

```
countProperties({a:1,b:2,c:3})  
// 3
```

Hint:

Use:

```
Object.keys()
```

7. Loop Through Object

Print all keys and values from this object.

```
const person = {  
  name: "Rahul",  
  age: 22,  
  city: "Delhi"  
}
```

Hint:

Use:

```
for...in
```

8. Check Property Exists

Check whether `"email"` exists inside an object or not.

Hint:

Use:

```
in
```

9. Merge Two Objects

Merge these two objects into one.

```
const obj1 = { a: 1, b: 2 }  
const obj2 = { c: 3, d: 4 }
```

Hint:

Use:

```
spread operator
```

10. Convert Object to Array

Convert this object into an array of key-value pairs.

```
const user = {  
  name: "Aman",  
  age: 21  
}
```

Hint:

Use:

```
Object.entries()
```

Problem Solving Level

11. Find Highest Value

Find the student with highest marks.

```
const marks = {  
  Anubhav: 95,  
  Rahul: 82,  
  Aman: 90  
}
```

Expected Output:

```
"Anubhav"
```

12. Sum of Object Values

Find total salary.

```
const salaries = {  
  john: 1000,  
  alex: 2000,  
}
```

```
    bob: 1500
  }
```

Expected Output:

```
4500
```

13. Nested Object Access

Print:

- city
- pincode

```
const user = {
  name: "Anubhav",
  address: {
    city: "Bhopal",
    pincode: 462001
  }
}
```

14. Object Method Practice

Create an object with:

- name
- marks
- method called `getResult`

If marks > 40:

```
"Pass"
```

else:

```
"Fail"
```

15. Convert Array to Object

Convert this array into an object.

```
const arr = ["name", "Anubhav", "age", 24]
```

Expected Output:

```
{  
  name: "Anubhav",  
  age: 24  
}
```

Harder Practice Questions

16. Frequency Counter

Count frequency of each character.

Input:

```
"banana"
```

Expected Output:

```
{  
  b: 1,  
  a: 3,  
  n: 2  
}
```

17. Group By Property

Group users by age.

```
const users = [  
  { name: "A", age: 20 },  
  { name: "B", age: 21 },  
]
```

```
[ { name: "C", age: 20 } ]
```

Expected Output:

```
{
  20: [
    { name: "A", age: 20 },
    { name: "C", age: 20 }
  ],
  21: [
    { name: "B", age: 21 }
  ]
}
```

18. Deep Property Check

Check whether this property exists:

```
"user.address.city"
```

inside an object dynamically.

Hint:

Use:

```
split(".")
```

19. Object Comparison

Check if two objects have same keys and values.

Example:

```
{a:1,b:2}
{a:1,b:2}
```

Expected Output:

true

20. Remove Duplicate Objects

Remove duplicate objects from array based on `id`.

```
[
  {id:1,name:"A"},
  {id:2,name:"B"},
  {id:1,name:"A"}
]
```

Expected Output:

```
[
  {id:1,name:"A"},
  {id:2,name:"B"}
]
```